



Wi-Fi 6 Routers

They're finally here

Wi-Fi 6 routers were announced by most brands around the CES / COMPUTEX 2018 launch window and even released in certain countries around the same time frame. Alas, we never saw those routers here in India for a really long time. In fact, it was around mid-2019 that Netgear launched the RAX80 in India and then we saw every other brand slowly come out of the shadows with their Wi-Fi 6 networking products. To be honest, the first movers released products based on the Wi-Fi 6 draft standard rather than on the final ratified standard and we can see how that makes sense from a manufacturer's perspective. And since the standard was ratified, we've seen Wi-Fi routers coming in at the mid-range and high-end price points. Wi-Fi 6 is designed for denser environments where you have a lot more clients but the problem of density of access points competing for the same airtime remained. That's being addressed with Wi-Fi 6e, the extension of the Wi-Fi 6 standard which makes a provision for a new 6 GHz band which is barely used by other radio standards. This would be the biggest change to the Wi-Fi protocol in 20 years and we should hopefully see devices using the new Wi-Fi 6e standard around this time next year.

WINNER: ASUS ROG RAPTURE GT-AX11000

ASUS is positioning the ROG Rapture GT-AX11000 as a Wi-Fi 6 gaming router. It's based on the Broadcom 4908 platform and has one 2.4 GHz band rated for 1148 Mbps and two 5 GHz bands which can handle 4804 Mbps each. There's one 2.5G Ethernet port aside from the usual Gigabit LAN ports so folks who have internet connections rated for more than 1G speeds now have a router that can actually handle that kind of traffic. LAN-to-LAN transfer speeds are limited to 1G and we ended up seeing 945 MBps for that. There are two USB 3.1 Gen 1 ports so you can have more external storage media attached for data sharing or for the download box. On a single



**ASUS ROG
Rapture
GT-AX11000**
Price: ₹36,950



D-Link DIR-X6060
Price: ₹26,899



Linksys Velop MX5300
Price: ₹32,199

client test we can see speeds up to 560 Mbps on the 5 GHz band which was the highest among all the routers we tested.

RUNNER UP AND BEST BUY: D-LINK DIR-X6060

Using the same Broadcom 4908 platform, the D-Link DIR-X6060 comes in second place this year. It's a dual-band Wi-Fi 6 router so the cumulative bandwidth among all clients is 6000 Mbps. While the difference between the cumulative bandwidths of the two top-contenders this year is massive, the single client performance isn't that far apart. We saw 524 Mbps on the 5 GHz band and on the 2.4 GHz band we'd see about 183 Mbps throughput. It comes with a single USB 3.1 Gen 1 port and has a 2.5G WAN port just like the ROG Rapture. The D-Link DIR-X9000 would have been a direct competitor to the Rapture but it is yet to be released this year. However, for the vast majority of users, either of the two routers should do just fine.

RUNNER UP: LINKSYS VELOP MX5300

Next in line is the Linksys MX5300 which is actually part of a mesh system but we ran the tests on a single node. It's a triband Wi-Fi router with one 2.4 GHz band and two 5 GHz bands. One of the 5 GHz bands even acts as a dedicated backhaul. There are only Gigabit ports on the MX5300 and all the antennas are internal so it's aesthetically pleasing but loses out on some performance. On the 5 GHz band we saw 446 Mbps of throughput and 2.4 GHz performance was in line with the rest of the contenders. So it's still a great router, especially in a mesh setup but among the high-end devices we tested, it lagged behind the rest.

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